C. System Administrator

time limit per test: 1 second
memory limit per test: 256 megabytes

Bob got a job as a system administrator in X corporation. His first task was to connect n servers with the help of m two-way direct connection so that it becomes possible to transmit data from one server to any other server via these connections. Each direct connection has to link two different servers, each pair of servers should have at most one direct connection. Y corporation, a business rival of X corporation, made Bob an offer that he couldn't refuse: Bob was asked to connect the servers in such a way, that when server with index v fails, the transmission of data between some other two servers becomes impossible, i.e. the system stops being connected. Help Bob connect the servers.

Input

The first input line contains 3 space-separated integer numbers n, m, v $(3 \le n \le 10^5, 0 \le m \le 10^5, 1 \le v \le n)$, n — amount of servers, m — amount of direct connections, v — index of the server that fails and leads to the failure of the whole system.

Output

If it is impossible to connect the servers in the required way, output `-1`. Otherwise output m lines with 2 numbers each — description of all the direct connections in the system. Each direct connection is described by two numbers — indexes of two servers, linked by this direct connection. The servers are numbered from 1. If the answer is not unique, output any.

Examples

input

```
5 6 3
```

output

```
1 2
2 3
3 4
4 5
1 3
3 5
```

input

```
6 100 1
```

output

```
-1
```

Copy Copy Copy Copy

Codeforces Beta Round 22 (Div. 2 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
371735456	Apr/19/2026 11:23	Accepted
371735162	Apr/19/2026 11:20	Accepted
371734892	Apr/19/2026 11:18	Wrong answer on test 1
371734637	Apr/19/2026 11:15	Wrong answer on test 3

→ Problem tags

graphs *1700

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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